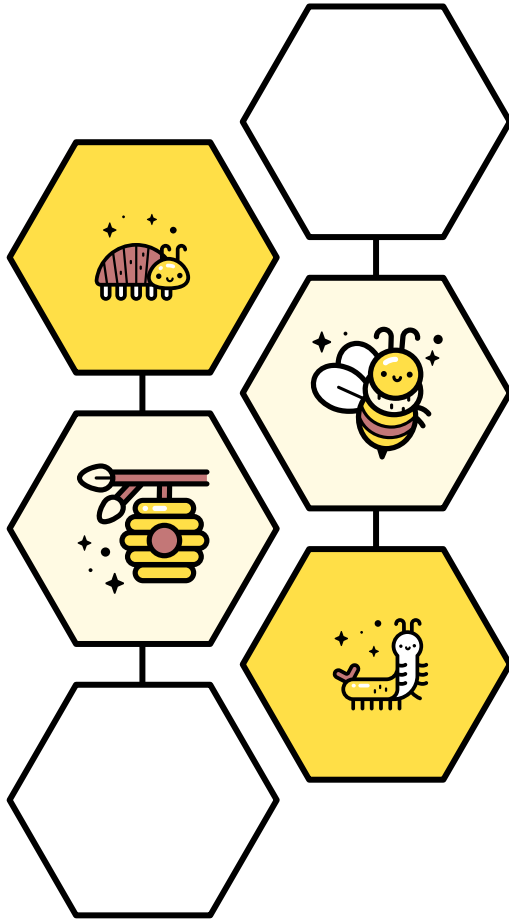




Hyperhive

**PERSONALIZES ADAPTIVE
LEARNING SYSTEM**

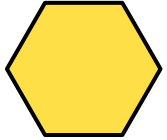
For Sekolah Sri Acmar



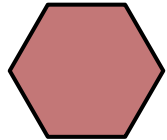
1.0 Introduction – Background



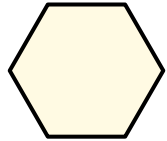
About Ever Core Labs (Company's Overview)



Project Undertaken (HyperHive)



HyperHive



1.0 Introduction – Background

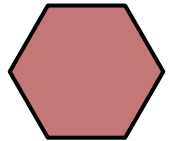


About Ever Core Labs (Company's Overview)

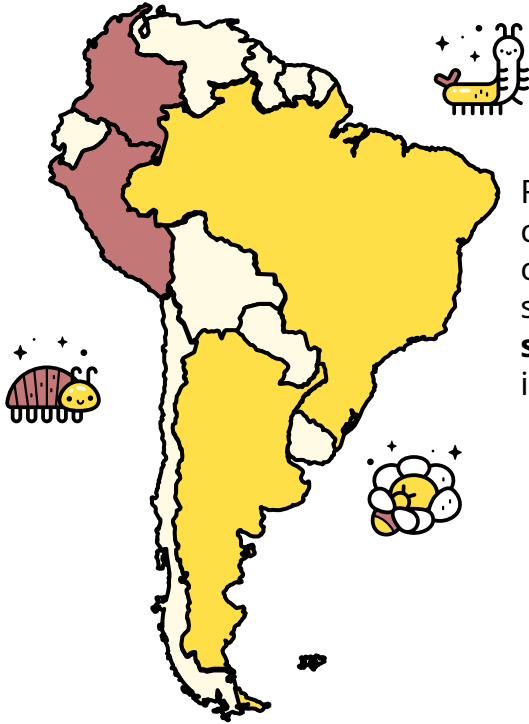
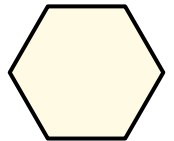
Founded **in 2020**, Ever Core Labs is a forward-thinking technology company headquartered in **Shah Alam, Selangor**. Ever Core Labs is currently focused on designing several groundbreaking projects, with a strong emphasis on **advancing student learning platforms and systems**. One of the main project Ever Core Labs currently focusing on is **HyperHive**; Personalized Adaptive learning System.



Project Undertaken (HyperHive)



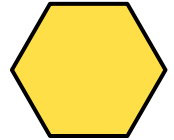
Hyperhive



1.0 Introduction – Background



About Ever Core Labs (Company's Overview)

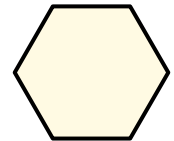


Project Undertaken (HyperHive)



The company is currently undertaking a project to design the **Personalized Adaptive Learning System (PALS), HyperHive**, a concept tailored specifically for *Sekolah Sri Acmar*, aiming to revolutionize the school's approach to education.

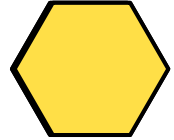
Hyperhive



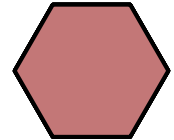
1.0 Introduction – Background



About Ever Core Labs (Company's Overview)



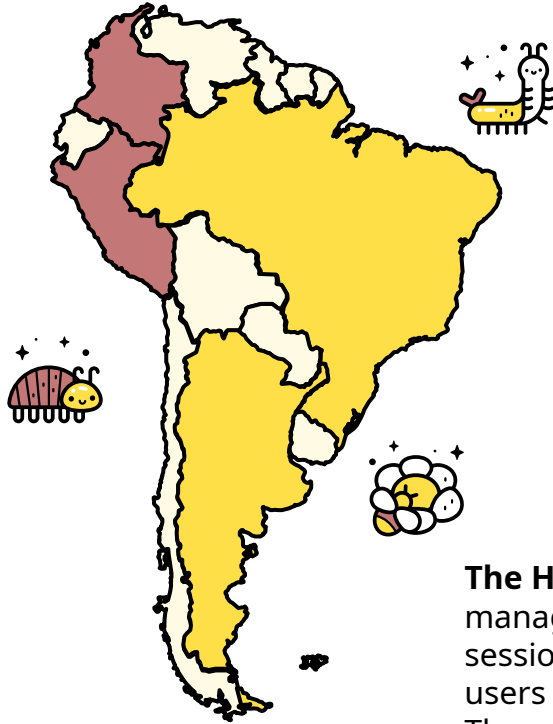
Project Undertaken (HyperHive)



Hyperhive



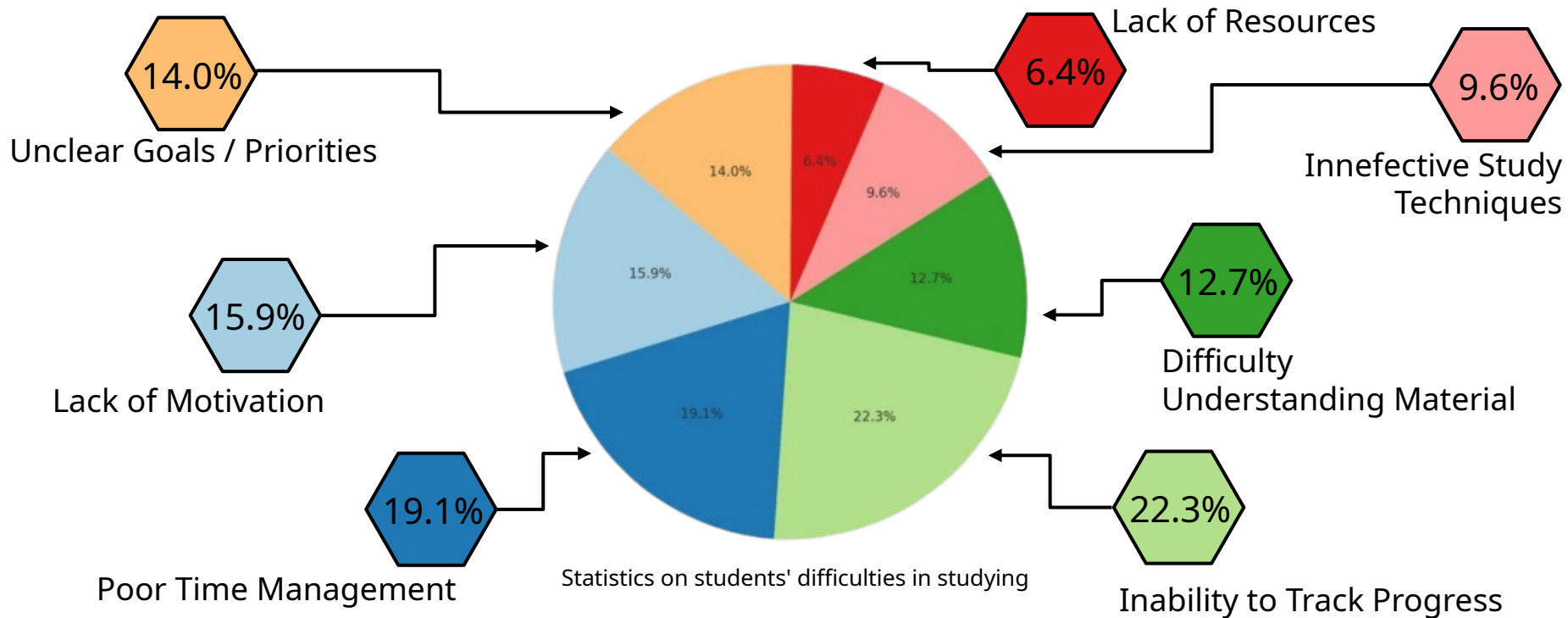
The HyperHive; Personalized Adaptive Learning System is a study management system designed to assist users in organizing their study sessions with the implementation of Pomodoro technique. This system allows users to **register, log in, manage subjects, and track their study progress**. The primary goal of HyperHive is to **enhance productivity** and learning efficiency by providing a structured approach to studying.





2.0 Problem Statement

According to the data gathered, **STEM students of Sekolah Sri Acmar struggle with varied learning paces, limited support, and poor time management.** This project aims to design a learning system for high school STEM students to address these challenges.



3.0 Objective

i. To analyze Personalized Adaptive Learning System requirements

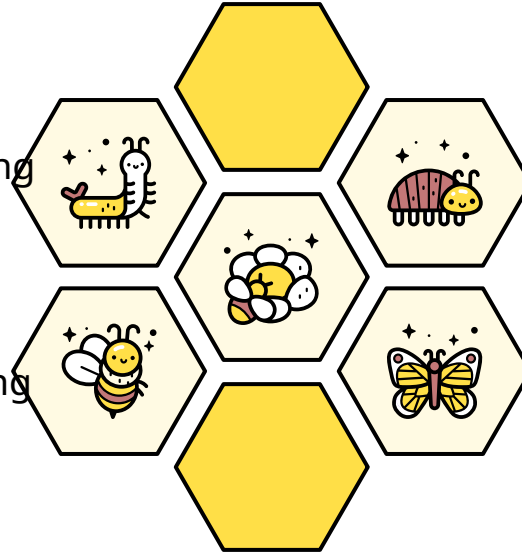
vi. To test algorithm of Personalized Adaptive Learning System using Flowgorithm

v. To see the expected screen of Personalized Adaptive Learning System

iv. To create pseudocodes for Personalized Adaptive Learning System

ii. To construct Personalized Adaptive Learning System modules

iii. To draw flowchart for the Personalized Adaptive Learning System





4.0 Methodology

Problem Analysis

01

02

Algorithm Design

Algorithm Implementation

03

04

Program Testing & Debugging

05

Program Documentation





4.0 Methodology

Personalized Adaptive Learning System (PALS) adheres to the Programming Development Life Cycle (PDLC) to ensure a systematic and structured approach to creating a robust and efficient system

The designing of PALS system is only up until the second phase of PDLC that is Algorithm Design and one of the fifth phase that is System Documentation

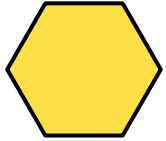


4.0 Methodology

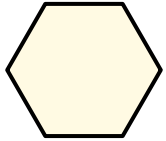


Problem Analysis

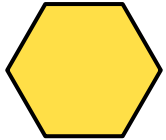
Problem analysis is **to analyze the problem that high school students in STEM faced**, and to come up a solution that works for them to ensure their learning journey goes smoothly. PALS identifies the need for an adaptive learning system that **personalizes study plans** and **enhances productivity** using suitable techniques according to their learning styles.



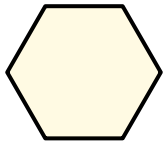
Algorithm Design



Algorithm Implementation

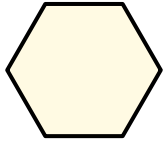


Program Testing & Debugging



Program Documentation

4.0 Methodology

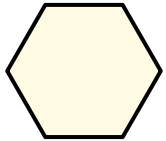


Problem Analysis

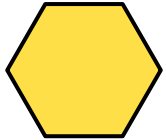


Algorithm Design

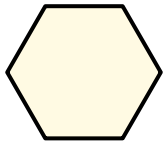
To visualize the problem where we focused on planning and creating the **module** or structure of a solution to the problem. **The main goal** is to break down the problem into sets of modules and efficiently solve it step by step.



Algorithm Implementation



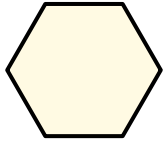
Program Testing & Debugging



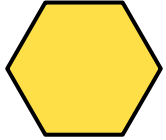
Program Documentation



4.0 Methodology



Problem Analysis

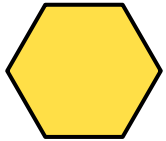


Algorithm Design

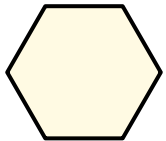


Algorithm Implementation

To transform the designed algorithm into a functional system using **c++ programming language** that automates tasks, optimizes processes, and adapts dynamically to meet the user needs effectively.



Program Testing & Debugging

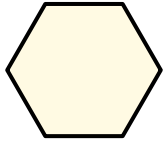


Program Documentation

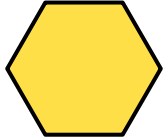




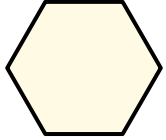
4.0 Methodology



Problem Analysis



Algorithm Design



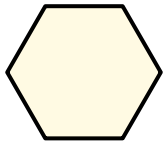
Algorithm Implementation



Program Testing & Debugging

To test the program which is the process of executing a program with the **intent of finding errors in the system.**

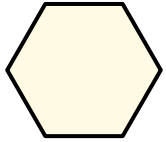
PALS **undergoes rigorous testing** to ensure **functionality, reliability, and usability.**



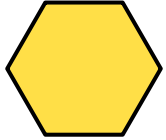
Program Documentation



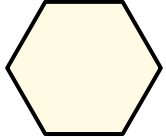
4.0 Methodology



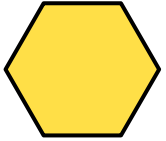
Problem Analysis



Algorithm Design



Algorithm Implementation



Program Testing & Debugging



Program Documentation

Provides **clear user guides**, **technical details**, and **maintenance logs** to support usability, development, and system evolution. PALS includes built-in mechanisms for updates and bug fixes, ensuring it remains scalable and adaptive to user needs.

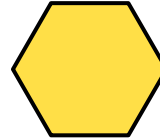
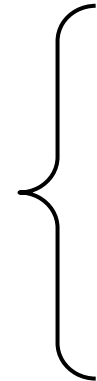
5.0 SOLUTIONS

Solution of the Personalized Adaptive Learning System (PALS)

The system solution of the Personalized Adaptive Learning System (PALS) refers to the comprehensive set of features, processes, and technologies designed to address the identified problems in **personalized learning and productivity**.

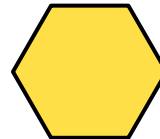


2 Main Examples of PALS potential solution :



1. User -Friendly Interface:

Design **an intuitive interface** that simplifies the scheduling and tracking of study sessions, making it accessible for all users



2. Algorithm Development:

Implement hyper-heuristics to **create adaptive study schedules** that respond to individual learning patterns and performance metrics.

5.0 SOLUTIONS

Solution of the Personalized Adaptive Learning System (PALS)

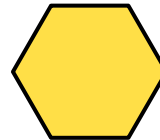
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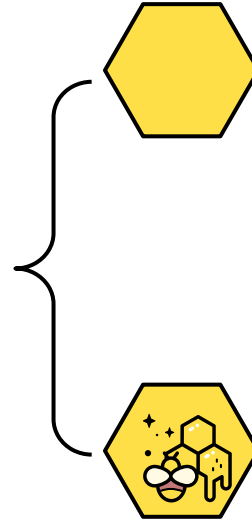


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5.1 Analyzing System Requirements



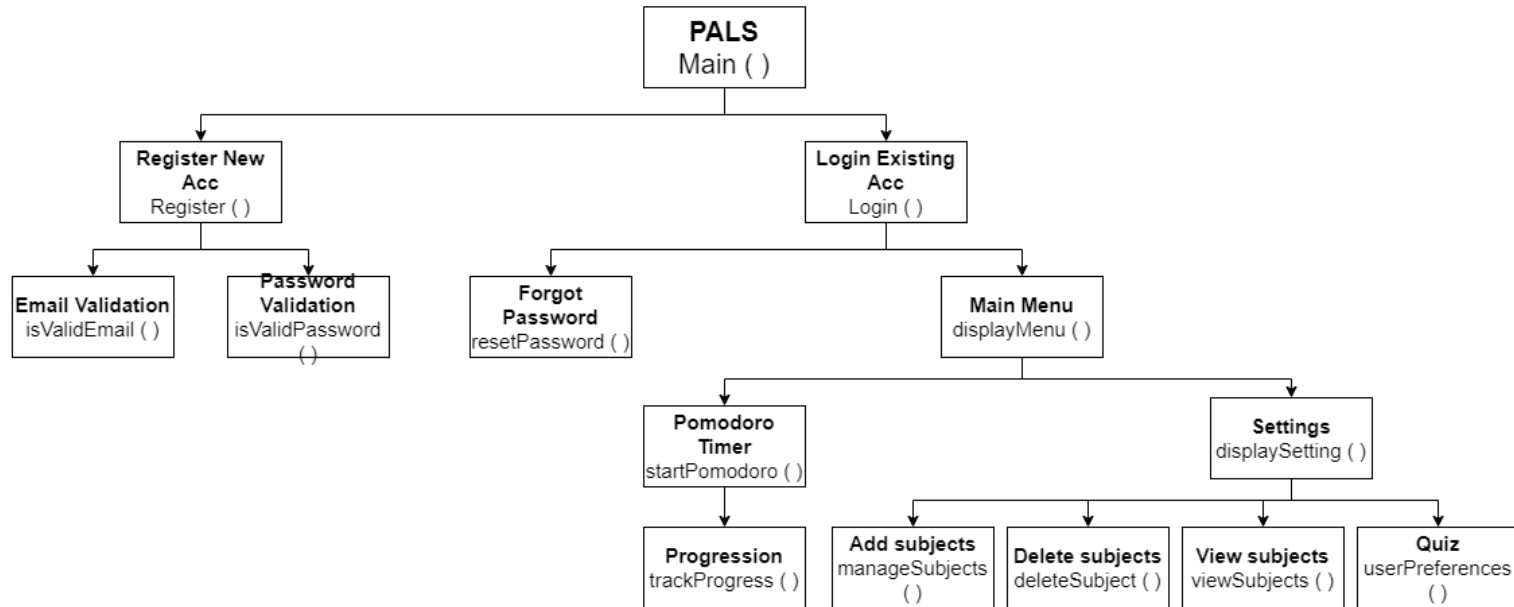
Analyzing system requirements is a critical step in the development of the Personalized Adaptive Learning System (PALS). Key insights were gathered during a **stakeholder meeting** with educators from Sekolah Sri Acmar, who provided valuable feedback based on their expertise and understanding of student needs. Their input emphasized in the list below :

- i. The system shall provide a Main Module that serves as the entry point and navigation hub for the system.
- ii. The system shall provide a Register Module that allows users to create a new account.
- iii. The system shall provide a Login Module that authenticates existing users to access the system.
- iv. The system shall provide a Forgot Password Module that enables users to securely reset their account password.
- v. The system shall provide a Main Menu Module that displays the primary navigation options for the user.
- vi. The system shall provide a Pomodoro Timer Module that lets users start and track their Pomodoro sessions.
- vii. The system shall provide a Progression Tracking Module that monitors and updates the user's study progress.
- viii. The system shall provide a Settings Module that allows users to customize their system preferences.
- ix. The system shall provide a Manage Subjects Module that enables users to add or edit study subjects.
- x. The system shall provide a Delete Subject Module that allows users to remove unwanted subjects.
- xi. The system shall provide a View Subjects Module that displays all active study subjects.
- xii. The system shall provide a User Preferences Module that lets users set and save personalized preferences

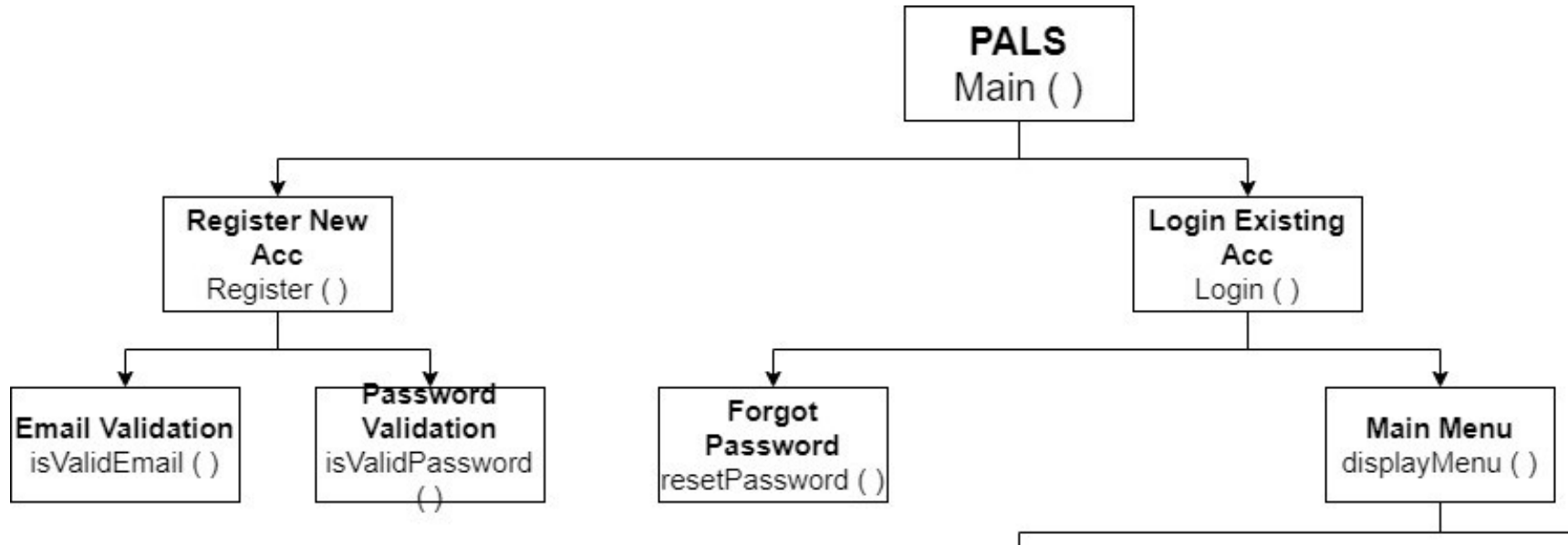
5.2 Designing Module



The System Module of Personalized Adaptive Learning System (PALS)

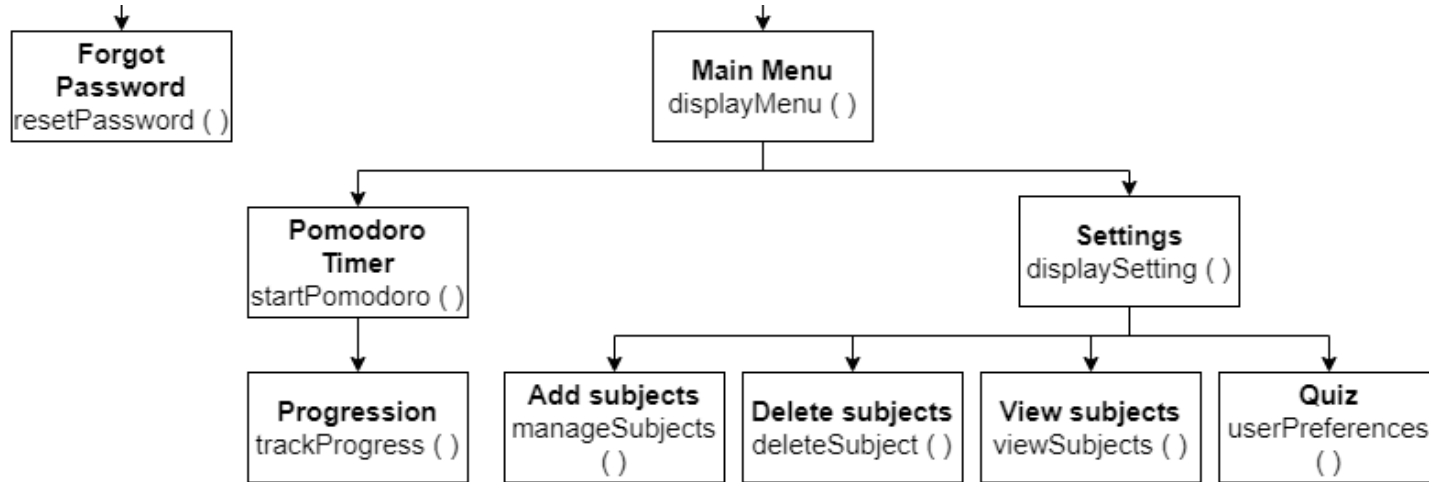


5.2 Designing Module



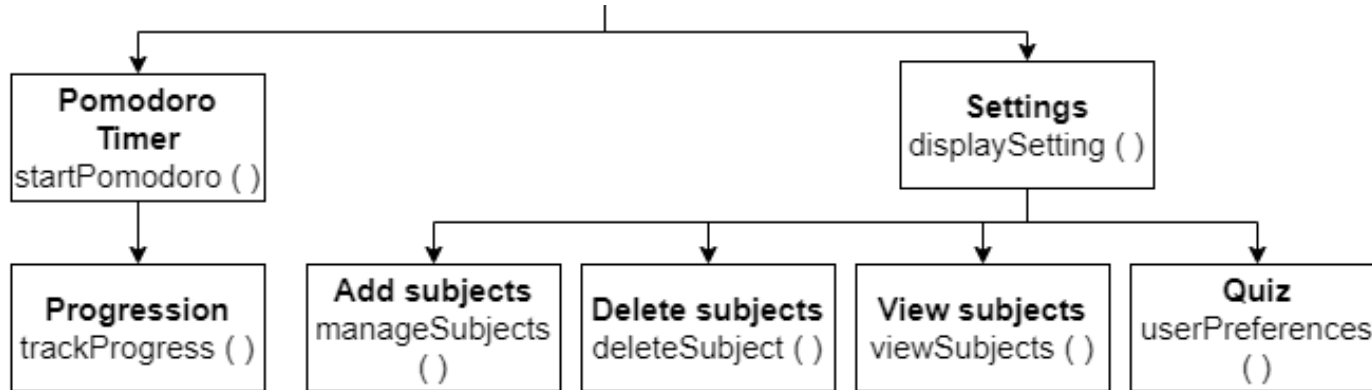
1. The Main Module serves as the entry point and navigation hub for the system.
2. The Register Module allows users to create a new account.
3. The Login Module authenticates existing users to access the system.
4. The Forgot Password Module enables users to securely reset their account password.
5. The Main Menu Module displays the primary navigation options for the user.

5.2 Designing Module



6. The Pomodoro Timer Module lets users start and track their Pomodoro sessions.
7. The Progression Module monitors and updates the user's study progress.
8. The Settings Module allows users to customize their system preferences.

5.2 Designing Module



9. The Manage Subjects Module enables users to add or edit study subjects.
10. The Delete Subject Module allows users to remove unwanted subjects.
11. The View Subjects Module displays all active study subjects.
12. The User Preferences Module lets users set and save personalized preferences

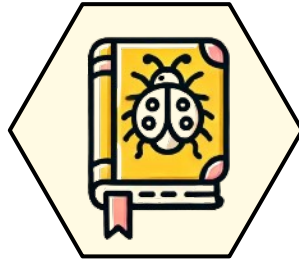
5.3 Formula / Calculation



Total Session

Learning Style Quiz

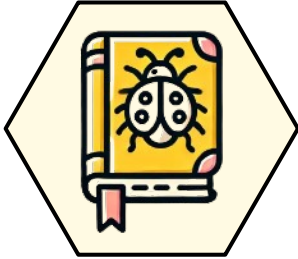
Pomodoro



5.3 Formula / Calculation



Total Session Done



Calculation / Formula for Total Session :

1. Read stored data for storedTotalSession into array
2. For every subjects, calculate total session done.
3. Calculate :

Using for loop for every subjects

`Subject[i]storedTotalSessions = storedTotalSessions + totalSessions`

Exp:

`MathsstoredTotalSessions = 8 + 5`

Updated:

`MathsstoredTotalSessions = 13`





5.3 Formula / Calculation

Learning Style Quiz Calculation

1. It will display questions, with three options A, B or C
2. Get answers for every question
3. Then calculate :

If every question's answer is mostly A

$A=A+1$

The answer will display its learnerType = "Visual learner "

If every question's answer are mostly B

$B=B+1$

The answer will display its learnerType = "Auditory learner"

If every question's answer are mostly C

$C=C+1$

The answer will display it's learnerType = "Kinesthetic learner"

Learning Style Quiz



5.3 Formula / Calculation



Pomodoro



Pomodoro Calculation / Formula :

- 1.It will retrieve number of subjects and subjects name from file
- 2.Get input from user for number of subjects that the user wants to learn
- 3.Get level of difficulty from user for each subject according to number of subjects
- 4.Get number of hours the user is going to learn that day
- 5.Then calculate :

```
pomodoroDuration = 25  
breakDuration = 5
```

```
sessionPerHour = 60 / ( pomodoroDuration + breakDuration )  
totalSessions = totalHours * sessionPerHour
```

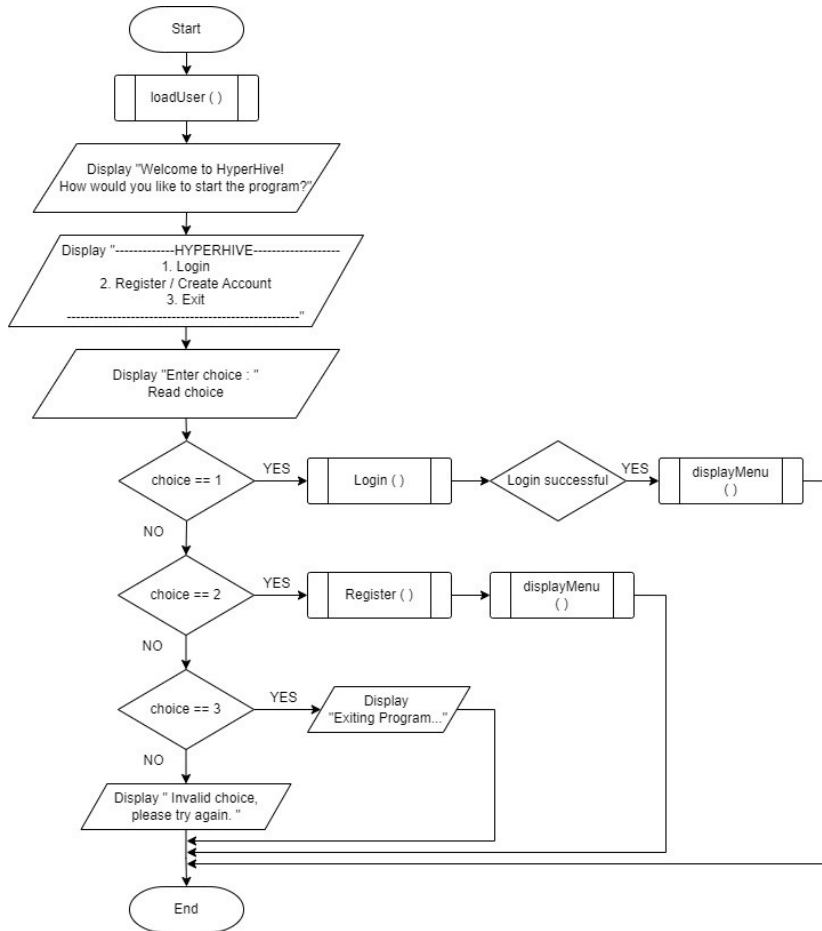
```
totalBreaktime = ( totalSessions - 1 ) * 5
```





5.4 Designing Flowchart





Main Module : Main()



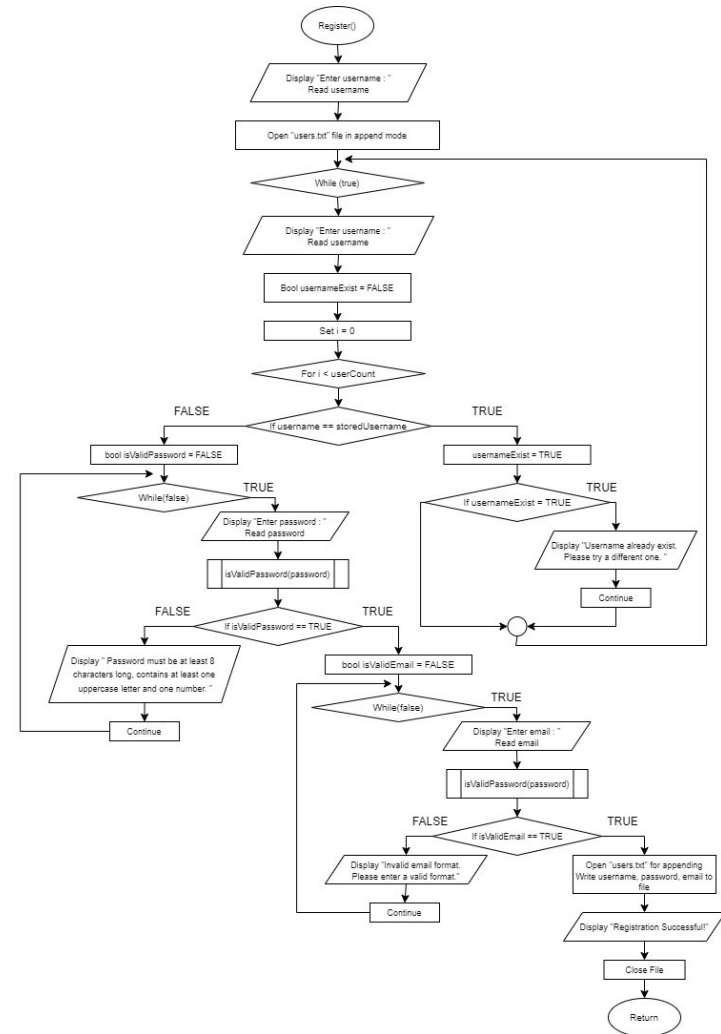
1. The program will call loadUsers() to load and retrieve data
2. The program will display welcome note and then prompt user with selections
3. Call of module or the next action will be taken based on selection chosen.

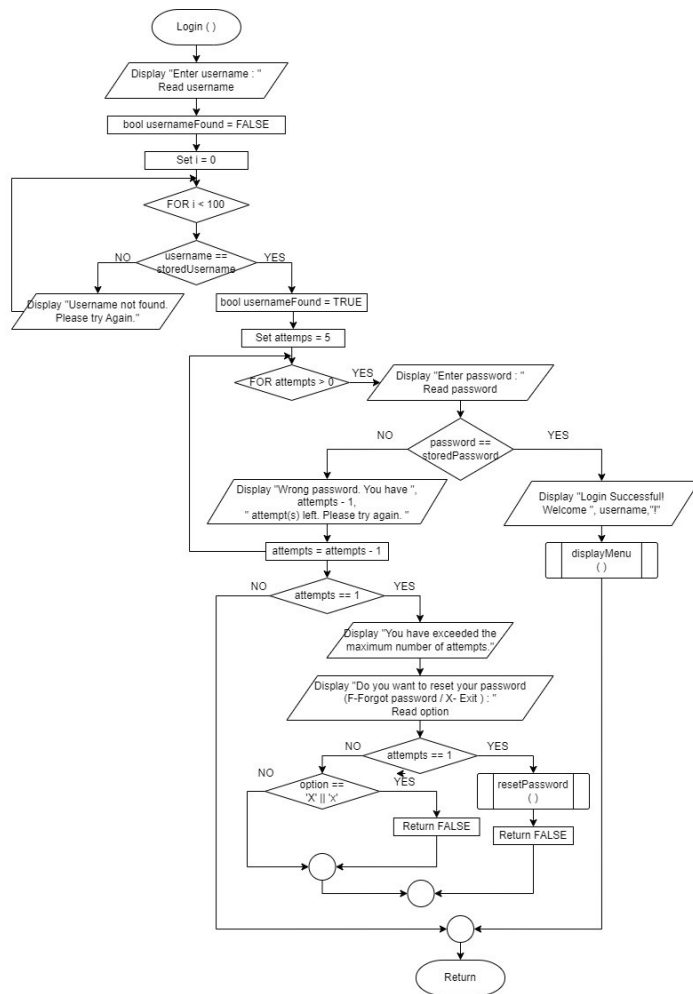




Register Module : Register()

1. The program will prompt user with enter username.
2. The program will open file "users.txt" to read and ensure username is yet to be written.
3. If username entered is yet taken, it will then prompt for password and email, and program will validate both password and email.
4. The data that was input will be written in the file sequentially.
5. The program will close file and return bool value back to Main().





Login Module : Login()



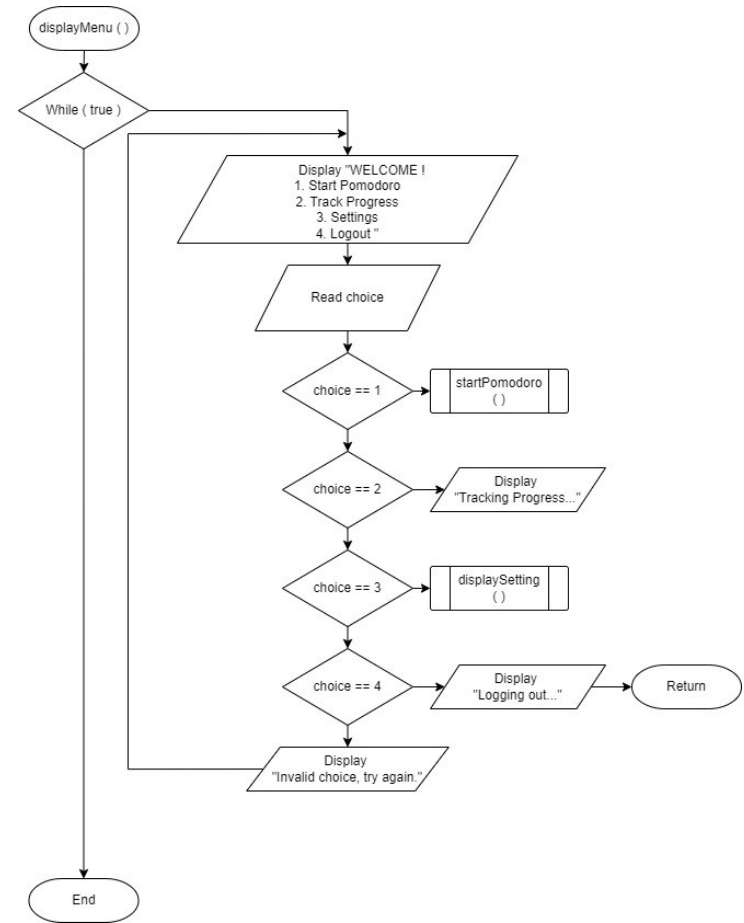
1. The program will prompt user with enter username.
2. The program will open file "users.txt" to read, skim through the file to check if the username existed.
3. If the username existed, program will then prompt user for password.
4. If password is the same as stored password in the file "users.txt" it will then call module for displayMenu().
5. If not, the program will then display wrong password, and prompt user again with password within 5 attempts.
6. If user still input wrong the password, it will then prompt user if they want to reset password.
7. If yes, it will call resetPassword. If no. it'll return to main() and return False value.





Display Menu Module : displayMenu()

1. While true, the program will provide and display selection for the user to take .
2. If first, it will call startPomodoro() module
3. If second, it will call display Tracking progress and call on trackProgress()
4. If third, it will call displaySettings().
5. If fourth, it will log out, and delete temporary data.



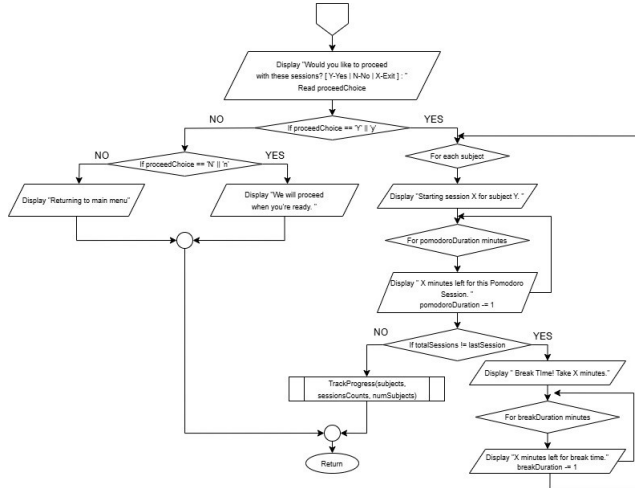
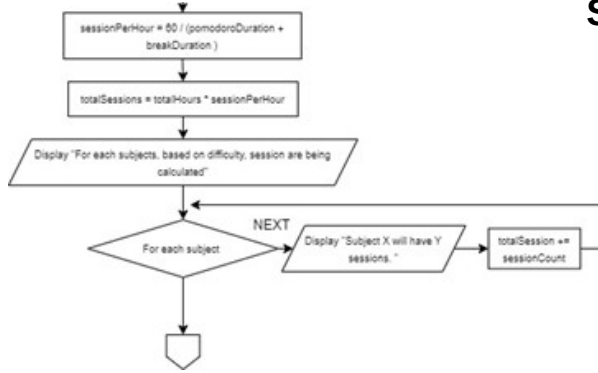
Start Pomodoro Module : startPomodoro() part 1



1. The program will open file "users.txt" for reading, to read subjects into array, then close the file.
2. It will then prompt user number of subjects they want to learn with max count from the subjects that was retrieve from the file.
3. Then it will display the available subjects as a reference for user to input.
4. Then using while loop, to get input for difficulty level for each subject chosen.
5. Program will ask user for the total hours they will like to learn for.



Start Pomodoro Module : startPomodoro() part 2 and part 3



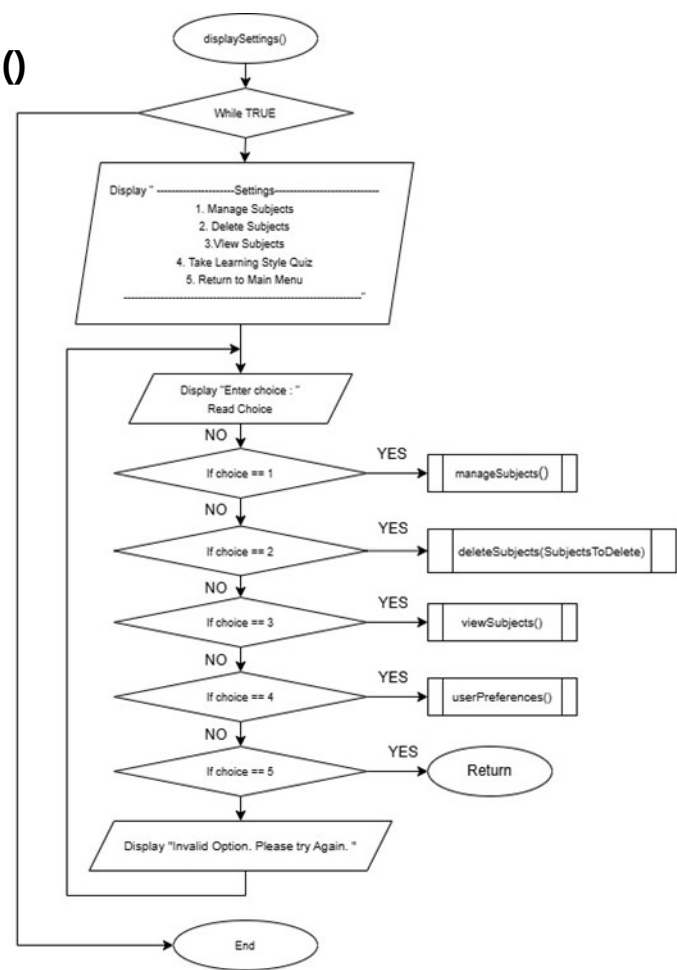
6. Program will then calculate the number of sessions for each subject based on it's level of difficulty.
7. Then it will use for loop, to display each subject calculated sessions.
8. Program will prompt user if they want to proceed with the pomodoro sessions.
9. If yes, pomodoro starts and continue with break time until last session is completed.
10. If no, it will display an appropriate message.
11. If exit, it will return to displayMenu().



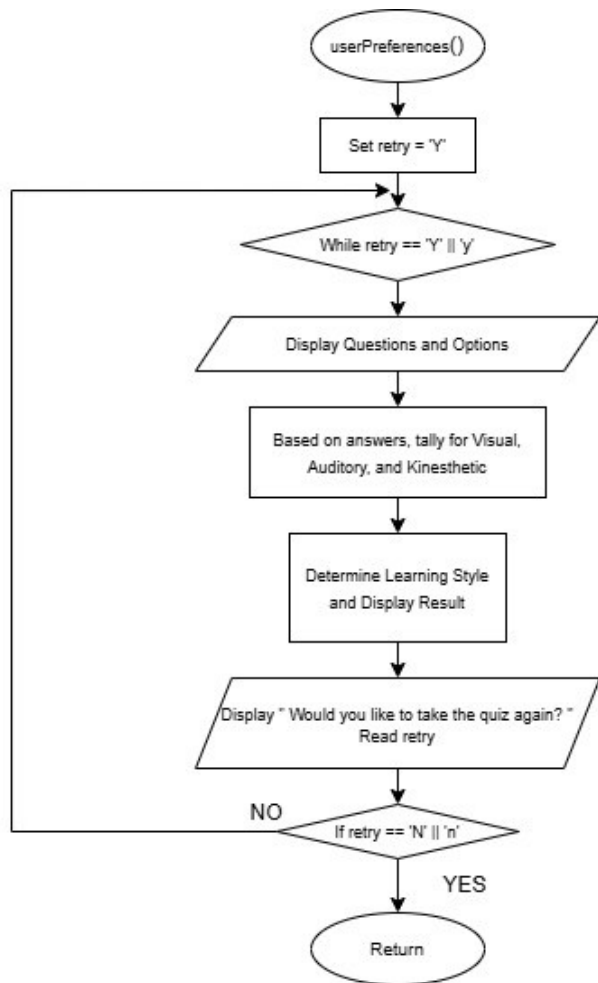


Display Settings module : displaySettings()

1. While true, the program will provide and display selection for the user to take .
2. If first, it will call manageSubjects() module,
3. If second, it will call deleteSubjects(subjectToDelete) module.
4. If third, it will call viewSubjects() module.
5. If fourth, it will call userPreferences() module.
6. If fifth, it will return to displayMenu().
7. If none the above, it will display " Invalid Option. Please Try Again"



User Preferences Module : userPreferences()



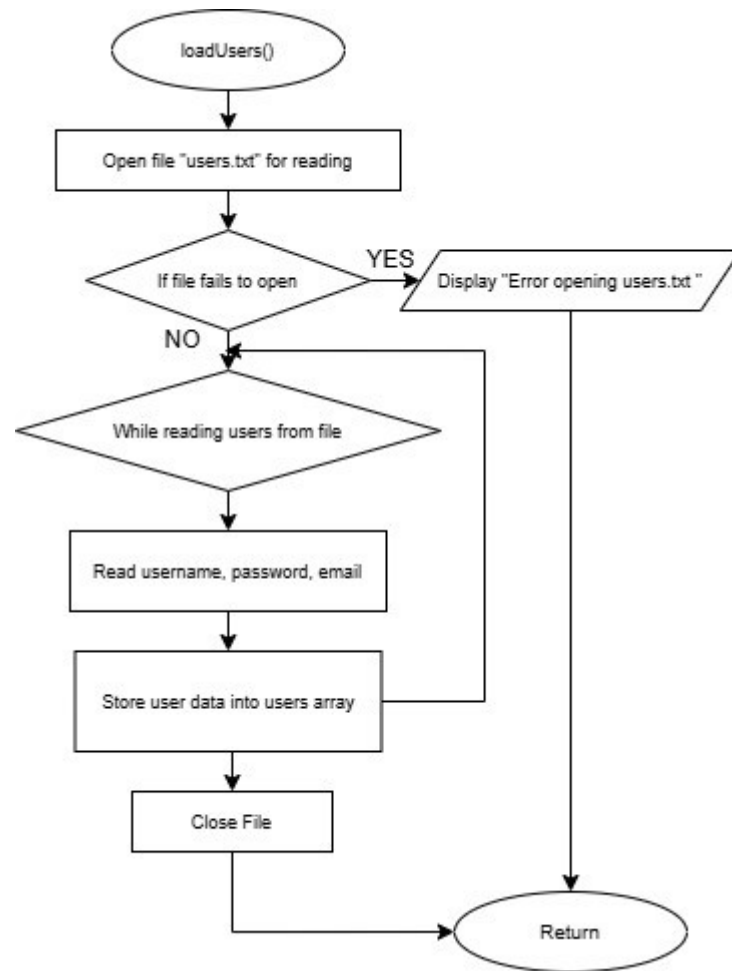
1. The program will set retry to Y
2. Using while loop, the program will display sets of questions and get input from user with 3 options that is A,B or C.
3. Based on the option, it will then calculate and determine what type of learner user is and display for user.
4. After done the program will prompt user if they want to retry, if yes it will redo and re-prompt the same questions.
5. If no, it will return to displaySettings().

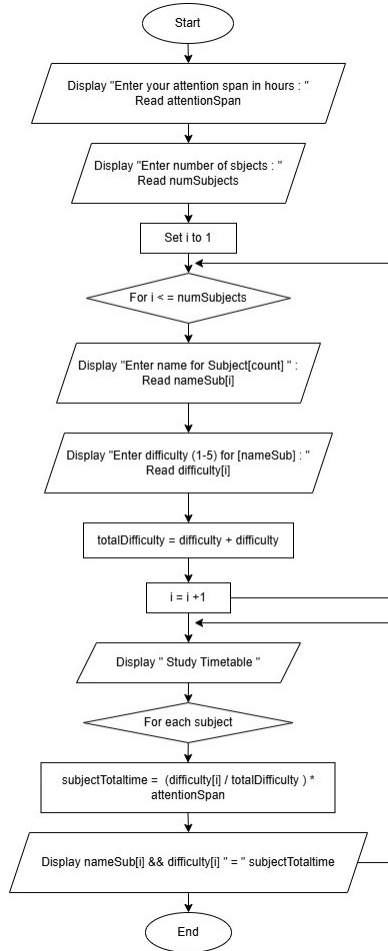




Load Users Data Module : loadUsers()

1. This module function is only for opening file for reading and read username, password and email and store them into array, then close file.
2. If file fails to open, it will display appropriate message.





Timetable : studyTime()



1. Program will prompt user for their attention span and number of subjects they want to learn.
2. For each subject, it will loop to enter name of the subjects and their difficulty.
3. It will then calculate total of hours user must spend for each subjects.
4. It then generates a timetable base on the calculation using for loop for each subject.





5.5 Pseudocode



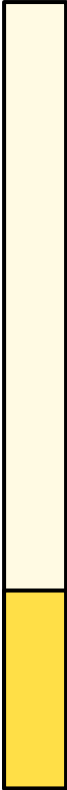
Pseudocode – Main()

```
Main()  
START  
  // Load user data from file  
  loadUsers()  
  
  // Display welcome message and options to the user  
  Display "Welcome to HyperHive! How would you like to start the program?"  
  
  While true  
    Display options:  
      1. Login  
      2. Register / Create Account  
      3. Exit  
  
    Read choice  
  
    If choice == 1:  
      If Login() is successful:  
        Display menu  
      Else:  
        Continue  
    Else If choice == 2:  
      Call Register()  
    Else If choice == 3:  
      Display "Exiting program."  
      Exit  
    Else:  
      Display "Invalid choice, please try again."  
  
End
```

Pseudocode – Register() part 1

```
Function Register()  
    // Get user input for username, password, and email  
    Display "Enter new username: "  
    Read username  
  
    While true:  
        Check if username already exists in users array  
        If exists:  
            Display "Username already exists. Please try a different one."  
            Continue  
        Else:  
            Break  
  
    While true:  
        Display "Enter new password: "  
        Read password  
        If password is not valid:  
            Display "Password must be at least 8 characters long, contain at least  
                one uppercase letter and one number."  
            Continue  
        Else:  
            Break
```


Pseudocode – Register() part 2



```
While true:
    Display "Enter your email: "
    Read email
    If email is invalid:
        Display "Invalid email format. Please enter a valid email."
        Continue
    Else:
        Break

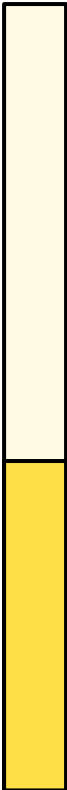
    Open "users.txt" for appending
    Write username, password, email to file
    Display "Registration successful!"
    Close file

End
```

Pseudocode – Login()

```
Function Login()
    // Get user input for username and password
    display "Enter your username: "
    read username
    For each user in users array:
        If username matches user data:
            ask for password
            For 5 attempts:
                display "Enter your password: "
                read password
                If password matches, return true
            Else:
                display "Wrong password. You have X attempts left."
                If last attempt, ask if user wants to reset password or exit
                display "You have exceeded the maximum number of attempts."
                display "Do you want to reset your password? (F - Forgot
                    Password / X - Exit): "
                read option
                If option = 'F' or option = 'f':
                    call ResetPassword(username)
                    return false
                Else If option = 'X' or option = 'x':
                    return false
            End for
        End if
    End for
    display "Username not found. Please try again."
    return false
End
```

Pseudocode – displayMenu()



```
Function displayMenu()  
    While true  
        Display menu options:  
            1. Start Pomodoro  
            2. Track Progress  
            3. Settings  
            4. Logout  
  
        Read choice  
  
        If choice = 1:  
            Call startPomodoro()  
        Else If choice = 2:  
            Display "Tracking progress..."  
        Else If choice = 3:  
            Call displaySettings()  
        Else If choice = 4:  
            Display "Logging out..."  
            Return  
        Else:  
            Display "Invalid choice, try again."  
    End while  
End
```

Pseudocode – startPomodoro() part 1

```
Function startPomodoro()
  open "subjects.txt" for reading // Load subjects from file
  read subjects into array
  close file
  display "How many subjects would you like to learn today? (Max X): "
  read numSubjects
  If numSubjects > available subjects:
    display "Limiting subjects to available subjects."
    numSubjects = available subjects
    display "Available subjects:"
  For each subject in subjects, display the subject and number
    For each subject:
      display "Rate the difficulty level from 1 to 5 for X: "
      read difficulty

      display "How many hours can you study today?"
      read totalHours

      const pomodoroDuration = 25 // minutes
      const breakDuration = 5 // minutes

      sessionsPerHour = 60 / (pomodoroDuration + breakDuration)
      totalSessions = totalHours * sessionsPerHour

      // Display subject sessions and calculate total sessions
      display "For each subject, based on difficulty, sessions are being
        calculated."
```

Pseudocode – startPomodoro() part 2

```
For each subject:
    display "Subject X will have Y sessions."
    totalSessions += calculated sessions

    display "Would you like to proceed with these sessions? Y-Yes |
        N-No | X-Exit: "
    read proceedChoice
    If proceedChoice = 'Y' or proceedChoice = 'y':
        For each subject:
            For each session:
                display "Starting Session X for subject Y."
                For pomodoroDuration minutes:
                    display "X minutes left for this Pomodoro
                        session."
                If not last session:
                    display "Break time! Take X minutes."
                    For breakDuration minutes:
                        display "X minutes left for break."
                    End for
            EndIF
        End for
    End for
```

Pseudocode – startPomodoro() part 3



```
trackProgress(subjects, sessionCounts, numSubjects)
    return
Else
    If( proceedChoice = 'N' or proceedChoice = 'n')
        display "We will proceed when you're ready."
        return
    Else
        If proceedChoice = 'X' or proceedChoice = 'x':
            display "Returning to main menu."
            Return
        EndIF
    EndIF
EndIF
End
```

Pseudocode – trackProgress()



```
Function trackProgress
Function trackProgress(subjects, sessionCounts, numSubjects)
    display "Tracking your progress:"
    For each subject, sessionCount:
        display "Subject: X - Sessions completed: Y"
    display "Great Job!"
End
```

Pseudocode – displaySettings()

```
Function displaySettings()
  While true
    Display settings options:
      1. Manage Subjects
      2. Delete Subject
      3. View Subjects
      4. Take Learning Style Quiz
      5. Return to Main Menu

    Read choice

    If choice = 1:
      Call manageSubjects()
    Else If choice = 2:
      Read subjectToDelete
      Call deleteSubject(subjectToDelete)
    Else If choice = 3:
      Call viewSubjects()
    Else If choice = 4:
      Call userPreferences()
    Else If choice = 5:
      Return
    Else:
      Display "Invalid option. Please try again."
  End while
End
```


Pseudocode – manageSubjects() and addSubject(newSubject)



```
Function manageSubjects()  
    Display "Enter new subject: "  
    Read newSubject  
    Call addSubject(newSubject)  
    Display "Subject added successfully."  
End
```

```
Function addSubject(newSubject)  
    Open "subjects.txt" for appending  
    Write newSubject to file  
    Close file  
End
```

Pseudocode – deleteSubject(subjectToDelete) and viewSubjects()



```
Function deleteSubject(subjectToDelete)
    Open "subjects.txt" for reading
    Open "temp.txt" for writing
    For each subject in subjects:
        If subject is not subjectToDelete, write to temp.txt
    Close files
    Replace "subjects.txt" with "temp.txt"
    Display success or failure message
End
```

```
Function viewSubjects()
    Open "subjects.txt" for reading
    Display list of subjects
    Close file
End
```

Pseudocode – userPreferences()



```
Function userPreferences()  
    // Initialize retry flag  
    retry = 'Y'  
    While retry = 'Y' or retry = 'y'  
        // Quiz to determine learning style  
        Display questions and options  
        Based on answers, tally counts for Visual, Auditory, and Kinesthetic  
        Determine learning style and display result  
        Display "Would you like to take the quiz again? (Y/N): "  
        Read retry  
    End while  
End
```

Pseudocode – loadUsers()



```
Function loadUsers()  
    Open "users.txt" for reading  
    If file fails to open:  
        Display "Error opening users.txt"  
        Return  
  
    While reading users from file:  
        Read username, password, email  
        Store user data into users array  
    Close file  
End
```

Pseudocode – isValidEmail(email) and isValidPassword(password)

```
Function isValidEmail(email)
    Find '@' in email
    Find '.' after '@'
    If '@' and '.' are present and '@' is before '.', return true
    Else return false
End
```

```
Function isValidPassword(password)
    If password length is less than 8, return false
    Check if password contains at least one uppercase and one digit
    If both are true, return true
    Else return false
End
```



5.6 Example of Expected Screen



HyperHive



STEM E-Learning for Students

Welcome to HyperHive! How would you like to start the program?



Login

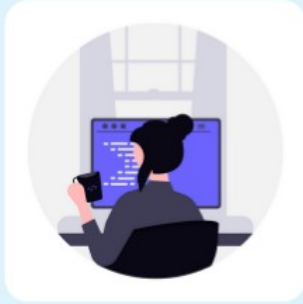
Register

Exit





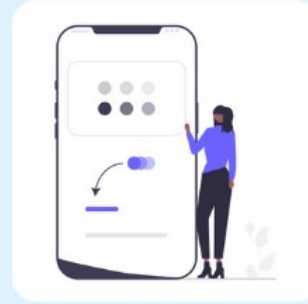
Welcome, username !



Start Pomodoro



Track Progress



Settings



Log Out



How many subjects would you like to learn today? (Max [subjectsCount]) :

3

Subjects Available
Maths
Biology
Algorithm

Rate the difficulty level from 1 to 5 for Maths :

5

Rate the difficulty level from 1 to 5 for Biology :

5

Rate the difficulty level from 1 to 5 for Algorithm :

5

How many hours can you study today? :

3

Done



Pomodoro Timer

Subject : Maths will have 2 sessions

Subject : Biology will have 2 sessions

Subject : Algorithm will have 2 sessions

Would you like to proceed with these sessions?

No

Yes

Exit

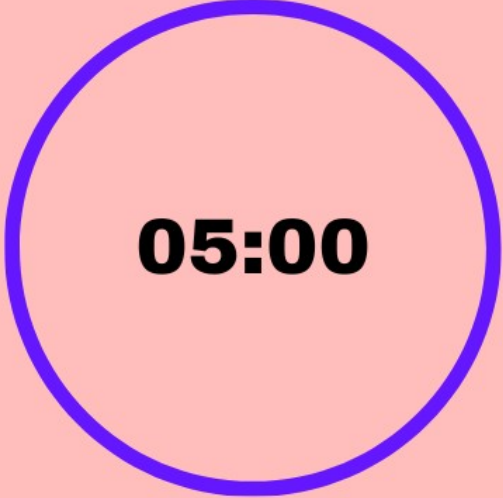
Starting Session 1 for Maths



25 : 00

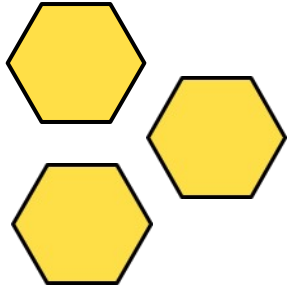
minutes left

Break Time starts Now! Take 5



05:00

minutes left



6.0 Outcome



6.1 Benefits of the System

The system enhances efficiency in learning by automating study session tracking, adaptive learning path generation, and progress analytics. It dynamically adjusts study plans based on performance data and provides curated content to suit different learning styles. Its streamlined operations ensure effective time management and consistent progress monitoring.

6.2 Solutions Provided by the System

The system simplifies learning management by automating content recommendations, performance analysis, and study session scheduling. It ensures flexibility through customizable settings and integration with adaptive algorithms, enabling real-time adjustments to study plans. Notifications and reminders optimize system functionality by maintaining consistent engagement and reducing missed sessions.

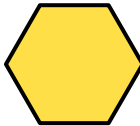
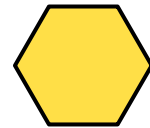


6.0 Outcome



6.3 Limitations of the System

The system relies heavily on stable internet connectivity and frequent updates, which may lead to technical issues during downtime. Compatibility challenges with diverse curriculum standards could limit its scalability in some regions. Additionally, high development and maintenance requirements may impact long-term cost-effectiveness for smaller institutions.

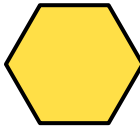
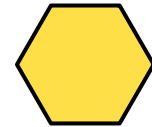


6.0 Outcome



Outcome for our HyperHive, Personalized Adaptive Learning System

The Personalized Adaptive Learning System is designed to optimize learning by automating key processes such as personalized study plan generation, progress tracking, and adaptive content recommendations. It aims to provide learners with a structured, efficient, and engaging study experience. The system is intended to enhance productivity, reduce learning gaps, and foster consistent academic growth, resulting in improved performance and better time management.



7.0 Conclusion



The HyperHive documentation serves as a detailed blueprint for future projects, reducing development time and enhancing code quality. Its focus on user-centric design helps create intuitive systems tailored to user preferences, while its adaptable principles allow expansion into new domains like training and wellness.

Insights from user feedback drive innovation and iterative improvements, supported by rapid prototyping and testing. The documentation also streamlines quality assurance, simplifies onboarding for new developers, and ensures efficient maintenance, making it a key resource for long-term growth and innovation.

